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The association between *Chlamydia pneumoniae* infection and prognosis in lung cancer patients: a prospective study

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Abstract

Background The prognostic value of *Chlamydia pneumoniae* (*Cpn*) infection in postoperative lung cancer patients remains unclear. This study aimed to evaluate the association between *Cpn* infection and survival in lung cancer patients.

Methods This study included 309 newly diagnosed primary lung cancer patients from three hospitals in Fuzhou, China. *Chlamydia pneumoniae* IgG and IgA antibodies were detected using a micro-immunofluorescence kit and a unified questionnaire was used to collect demographic and clinical information. Follow-up was conducted every six months until August 25, 2022, or death. Patients were followed up for a median duration of 74 months. Overall survival was compared between *Cpn* IgA and IgG positive/negative patients. Statistical analysis included the χ^2 test, Kaplan-Meier method, Log-rank test, and Cox proportional hazards regression model.

Results *Cpn* IgA infection status was found to be associated with lymph node metastasis in lung cancer patients. The overall survival of patients with *Cpn* IgA(++) was shorter compared to negative and positive patients (with IgA(-) group as the control group, $P < 0.001$, HR = 5.441, 95%CI: 2.341–12.645). Serum IgA positivity indicated chronic infection and was associated with poor prognosis. Multivariate analysis confirmed that *Cpn* IgA(++) infection was an independent risk prognostic factor for operable lung cancer patients (with IgA(-) group as the control group, HR = 3.39, 95%CI: 1.41–8.17).

Conclusion This study suggests that *Cpn* infection is an independent prognostic factor in lung cancer patients. Further research is needed to understand the underlying mechanisms and explore the potential of *Cpn* as a biomarker or therapeutic target for lung cancer.

Keywords *Chlamydia pneumoniae*, Lung cancer, Prognosis, Infection

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Background

Lung cancer (LC) is one of the most prevalent and deadliest forms of cancer worldwide. It's reported that LC was the second most commonly diagnosed cancer and the leading cause of cancer death, with an estimated 2.2 million new cases and 1.8 million deaths in 2020, representing approximately one in 10 (11.4%) cancers diagnosed and one in 5 (18.0%) deaths [1]. Low-dose Computed Tomography (LDCT) screening helps to improve the early diagnosis rate of lung cancer and prolong the survival time [2, 3]. However, the implementation of lung cancer screening remains a challenge due to the lack of obvious symptoms and the limitations of current clinical diagnostic techniques. The 5-year survival rate of patients with lung cancer is only 10–20% in most countries among those diagnosed during 2010 through 2014 [4]. The known prognostic factors of lung cancer include age, gender, smoking status, treatment, tumor size, tumor histological type, tumor metastasis, etc [5–8]. However, patients even with the same clinical characteristics might have dramatically different prognosis. The identification of new prognostic factors is of great importance to improve outcome prediction and tailor individualized treatment in patients with lung cancer.

It was reported that the percent of cancers associated with pathogenic microorganisms is estimated at 13% in 2018 [9]. Many pathogenic microorganisms have been proved to be related to the occurrence and development of lung cancer and might affect the prognosis of lung cancer, such as Hepatitis B virus, *Mycobacterium tuberculosis* and human papilloma virus [10–16]. *Chlamydia pneumoniae* (*Cpn*) is an obligate intracellular pathogen transmitted via droplets. It is a common cause of human respiratory diseases and most commonly manifests as pneumonia and bronchitis, and with the deepening research of *Cpn*, it is gradually found to be associated with chronic diseases such as cardiovascular disease, cerebral infarction and lung cancer [17–21]. *Cpn* infection can induce cellular and humoral immunity, and specific antibodies can be detected in the serum of patients. Usually serum immunoglobulin G (IgG) antibodies positive indicates that the body has previously been infected with *Cpn*, while serum immunoglobulin A (IgA) antibodies positive indicates the presence of *Cpn* repeated infection or chronic infection. Because *Cpn* can cause chronic infection, some researchers have found that *Cpn* is associated with the prognosis of coronary heart disease [22].

Nevertheless, there is no related research about the relationship of *Cpn* and the survival of patients with lung cancer currently. Therefore, this study analyzes the correlation between *Cpn* chronic and previous infection and lung cancer prognosis, aim to providing a scientific basis for the treatment and prognosis analysis of lung cancer.

Methods

Study subjects

The cases were from the Department of Thoracic Surgery and Respiratory Medicine of the First Affiliated Hospital of Fujian Medical University (Fujian, China), the Affiliated Union Hospital of Fujian Medical University (Fujian, China), and the 900th Hospital of Joint Logistics Support Force of the Chinese People's Liberation Army. (Fujian, China). A total of 309 patients with lung cancer were finally included.

The inclusion criteria for the cases were as follows: (1) patients who had not received any lung cancer-related treatment before, and were diagnosed as newly diagnosed primary lung cancer by surgical pathology; (2) the diagnosis date was from January 2011 to December 2017; (3) preoperative serum samples were collected for testing chlamydia infection status; (4) voluntarily participated and signed informed consent. The exclusion criteria for the cases were as follows: (1) patients who were diagnosed as secondary lung cancer by pathology; (2) patients who had a history of other cancers; (3) patients who were critically ill and could not answer questions clearly.

Information collection

A unified questionnaire was used, and detailed instructions and precautions were formulated. According to the principle of informed consent, trained investigators conducted interviews with all study subjects. The survey content included general information such as age, gender, height, weight at the time of the survey, smoking history, drinking history, past medical history and other lifestyle and health information, the additional file shows this in more detail [see Additional file 1]. And collected information on lung cancer diagnosis site, diagnosis time, histopathological type and other case information. Histopathological type were divided into three groups: adenocarcinoma, squamous cell carcinoma and others (including large cell carcinoma, carcinoid tumor, mucoepidermoid carcinoma and small-cell carcinoma). Whether lymph node metastasis has occurred was determined based on the pathological results of lymph node dissection during surgery.

Collection and storage of serum

After signing the informed consent form, 5 ml of fasting peripheral venous blood was collected from all study subjects using a non-anticoagulant vacuum blood collection tube at their first hospitalization. The collected samples were processed immediately, centrifuged at 2000 rpm for 10 min, and the separated serum was stored in a -80 °C freezer for future use, avoiding repeated freezing and thawing.

Detection of *Chlamydia pneumoniae* IgA and IgG antibodies in serum samples

micro-immunofluorescence (MIF) kit (Chlamydia IgG SeroFIA kit and Chlamydia IgA SeroFIA kit, DADE Behring, Savyon Diagnostics, Israel) were used to detect specific antibodies IgG and IgA against *Chlamydia pneumoniae* in serum. The operation was performed according to the instructions of the kit. The results were interpreted as follows: negative results [IgG(-) or IgA(-)] were no fluorescence staining or only dim background; positive results [IgG(+) or IgA(+)] were moderate diffuse apple green fluorescence staining; strongly positive results [IgG(++) or IgA(++)] were intense dazzling apple green fluorescence staining.

Follow-up

All study subjects were followed up every six months by trained investigators through telephone follow-up and electronic medical record review until August 25, 2022, or until lung cancer-related death. For patients who did not die from lung cancer, their deaths were recorded as censored data. Overall survival (OS) was defined as the time from the diagnosis of lung cancer to death or the last follow-up.

Statistical analysis

The χ^2 test was used to test the balance of baseline characteristics among different groups of *Chlamydia pneumoniae* infection. The Kaplan-Meier method was used to calculate the mean survival time, survival rate and plot survival curves. The Log-rank test was used to compare survival rate differences among groups. The Cox proportional hazards regression model was used to analyze potential prognostic factors. All analyses were performed by SPSS 24.0 software package (IBM Corporation, Armonk, New York, USA). All P values were based on a two-sided test with an α of 0.05.

Results

Characteristics of lung cancer patients with different *Chlamydia pneumoniae* infection status

Among the 309 lung cancer patients included, 90 (29.1%) were *Cpn*-IgG(-), and the number of IgG(+) and IgG(++) cases were 191 (61.8%) and 28 (9.1%), respectively. There were no significant differences among different *Cpn* serum IgG groups in terms of gender, age, BMI, smoking history, drinking history, pathological type, tumor maximum diameter, lymph node metastasis and chemotherapy.

The number of cases with IgA(-), IgA(+) and IgA(++) were 180 (58.2%), 121 (31.2%) and 8 (2.6%), respectively. There were no significant differences among the groups in terms of gender, age, smoking history, drinking history, pathological type, tumor maximum diameter and

chemotherapy. However, the lymph node metastasis rate of the IgA(++) group was higher than that of the IgA(-) and IgA(+) groups, and there were also statistical differences in the distribution of BMI among the three groups. (Table 1)

Clinical characteristics and prognosis of lung cancer

Univariate survival analysis showed that smoking history, drinking history, case type, tumor maximum diameter, lymph node metastasis and *Chlamydia pneumoniae* serum IgA infection status were associated with the overall survival of lung cancer patients. (Table 2)

The KM survival curves of different *Chlamydia pneumoniae* serum IgA and IgG infection status are shown in Fig. 1. It can be seen that there is a similar association between *Chlamydia pneumoniae* serum IgA and IgG infection and the OS of lung cancer patients. There is little difference in survival between IgA or IgG positive and negative patients, but lung cancer patients with strongly positive infection have a lower OS than the other two groups. In different *Chlamydia pneumoniae* serum IgA infection groups, the Log-rank test showed that this prognostic difference was statistically significant. (Fig. 1)

Including the indicators with significant statistical differences in univariate survival analysis into the multivariate Cox regression model and using the Backward: LR method for analysis, it was found that smoking (HR=1.55, 95%CI: 1.05–2.31), lymph node metastasis (HR=3.40, 95%CI: 2.27–5.08), larger tumor maximum diameter (with tumor maximum diameter ≤ 3 cm as the reference group, HR=1.54 for 3~7 cm group, 95%CI: 1.02–2.32; HR=2.36 for >7 cm group, 95%CI: 1.11–5.01), and *Chlamydia pneumoniae* serum IgA (++) (with IgA(-) group as the control group, HR=3.39, 95%CI: 1.41–8.17) can all associated with a decrease in the overall survival of lung cancer patients. (Table 3)

Discussion

Cpn infection has been confirmed to be closely related to the occurrence of lung cancer, which was also confirmed by our previous study [23]. However, the prognostic value of *Cpn* infection for the survival of postoperative patients with lung cancer has not been evaluated. To our knowledge, this study is the first to combine *Cpn* infection with prognosis in patients with lung cancer.

In our study, we found that *Cpn* IgA infection status was associated with lymph node metastasis in lung cancer patients, suggesting that *Cpn* infection may play an important role in the progression of lung cancer. The overall survival of *Cpn* IgG and IgA strong positive patients was shorter than that of negative and positive patients, and this situation was significantly different in different *Chlamydia* IgA infection status. Serum IgA positive usually implies chronic infection, suggesting that

Table 1 Clinicopathologic characteristics in 309 patients with lung cancer

Prognostic factor	Patients (%) n = 309	Cpn IgG(-) n = 90	Cpn IgG(+) n = 191	Cpn IgG(++) n = 28	χ^2	P	Cpn IgA(-) n = 180	Cpn IgA(+) n = 121	Cpn IgA(++) n = 8	χ^2	P
Gender					2.953	0.228					
Male	181(58.6)	48(53.3)	113(59.2)	20(71.4)			99(55.0)	75(62.0)	7(87.5)	4.112	0.116 ^a
Female	128(41.4)	42(46.7)	78(40.8)	8(28.6)			81(45.0)	46(38.0)	1(12.5)		
Age(years)					0.451	0.798					
<60	163(52.8)	50(55.6)	98(51.3)	15(53.6)			104(57.8)	54(44.6)	5(62.5)	5.324	0.067 ^a
≥ 60	146(47.2)	40(44.4)	93(48.7)	13(46.4)			76(42.2)	67(55.4)	3(37.5)		
BMI(kg/m²)					2.21	0.697				11.237	0.017 ^a
<18.5	20(6.5)	5(5.6)	12(6.3)	3(10.7)			10(5.6)	8(6.6)	2(25.0)		
18.5 ~ 24	184(59.5)	51(56.7)	115(60.2)	18(64.3)			99(55.0)	79(65.3)	6(75.0)		
≥ 24	105(34.0)	34(37.8)	64(33.5)	7(25.0)			71(39.4)	34(28.1)	0(0)		
Smoking					2.178	0.337				2.759	0.240 ^a
Never	151(48.9)	46(51.1)	95(49.7)	10(35.7)			95(52.8)	53(43.8)	3(37.5)		
Ever (former + current)	158(51.1)	44(48.9)	96(50.3)	18(64.3)			85(47.2)	68(56.2)	5(62.5)		
Alcohol					3.471	0.176				5.721	0.057
Never	225(72.8)	71(78.9)	132(69.1)	22(78.6)			140(77.8)	79(65.3)	6(75.0)		
Ever (former + current)	84(27.2)	19(21.1)	59(30.9)	6(21.4)			40(22.2)	42(34.7)	2(25.0)		
Pathological type					2.289	0.683				6.088	0.154 ^a
Adenocarcinoma	230(74.4)	67(74.4)	143(74.9)	20(71.4)			133(73.9)	93(76.9)	4(50.0)		
Squamous cell carcinoma	58(18.8)	15(16.7)	38(19.9)	5(17.9)			33(18.3)	23(19.0)	2(25.0)		
others ^b	21(6.8)	8(8.9)	10(5.2)	3(10.7)			14(7.8)	5(4.1)	2(25.0)		
Tumor maximum diameter(cm)					2.118	0.718 ^a				6.624	0.139 ^a
≤ 3	172(55.7)	50(55.6)	109(57.1)	13(46.4)			109(60.6)	60(49.6)	3(37.5)		
3 ~ 7	122(39.5)	37(41.1)	72(37.7)	13(46.4)			62(34.4)	56(46.3)	4(50.0)		
>7	15(4.9)	3(3.3)	10(5.2)	2(7.1)			9(5.0)	5(4.1)	1(12.5)		
Lymph node metastasis					3.868	0.145				9.572	0.008
Yes	90(29.1)	32(35.6)	48(25.1)	10(35.7)			46(25.6)	38(31.4)	6(75.0)		
No	219(70.9)	58(64.4)	143(74.9)	18(64.3)			134(74.4)	83(68.6)	2(25.0)		
Chemotherapy					2.413	0.299				0.391	0.809 ^a
Yes	174(56.3)	56(62.2)	101(52.9)	17(60.7)			99(55.0)	70(57.9)	5(62.5)		
No	135(43.7)	34(37.8)	90(47.1)	11(39.3)			81(45.0)	51(42.1)	3(37.5)		

^a: Using Fisher's exact test; ^b: Including large cell carcinoma, carcinoid tumor, mucoepidermoid carcinoma and small-cell carcinoma

chronic *Cpn* infection is associated with poor prognosis in lung cancer patients. Multivariate survival analysis also confirmed that *Cpn* IgA(++) infection is an independent risk prognostic factor for operable lung cancer patients, and the risk of death is 3.39 times higher than that of *Cpn* IgA(-) patients (95%CI: 1.41–8.17). In contrast, the presence of IgG antibodies did not show a significant difference in prognosis, which may be due to IgG indicating past infection rather than ongoing or repeated infection. This highlights the importance of IgA as a marker in assessing the prognosis of lung cancer patients with *Cpn* infection.

Chronic *Cpn* infection may affect the occurrence, development and prognosis of lung cancer through several ways. One potential mechanism by which *Chlamydia pneumoniae* infection may affect the prognosis of lung cancer patients is through the activation of inflammatory pathways. Inflammation plays a crucial role in tumor

development and progression, and chronic infections can contribute to a pro-inflammatory microenvironment that promotes tumor growth and metastasis. *Cpn* infection activates various inflammatory mediators, such as cytokines, chemokines, prostaglandins, and reactive oxygen species (ROS), that can cause DNA damage, epigenetic alterations, cell proliferation, angiogenesis, invasion, and metastasis [24–26]. Moreover, *Chlamydia trachomatis* (*Ctr*) can induce epithelial-mesenchymal transition (EMT), a process that confers stem-like properties and resistance to apoptosis and therapy on cancer cells [27, 28], *Cpn* may act a similar part in the occurrence and progress of lung carcinoma.

Another possible mechanism is that *Cpn* infection interferes with the signaling pathways that regulate cell growth and survival in lung cancer cells. *Cpn* infection can activate the nuclear factor-kappa B (NF-κB) pathway, which is a key transcription factor that regulates the

Table 2 Univariate survival analysis for OS in patients with lung cancer

Prognostic factor	mean survival time (months)	three-year survival rate	five-year survival rate	P
Gender				0.111
Male	95.4	76.6	67.3	
Female	103	80.2	74.9	
Age				0.316
<60	101.1	79	72.5	
≥ 60	94.3	77.1	68.1	
BMI				0.114
<18.5	83.6	63.3	58.1	
18.5~24	94.6	75.9	68.3	
≥ 24	106.7	84.7	76.6	
Smoking				0.006
Never	105.9	82.6	78.1	
Ever (former + current)	91.3	73.8	63.2	
Alcohol				0.002
Never	105.2	82	73.7	
Ever (former + current)	82.5	67.8	61.7	
Cpn IgG Status				0.363
Cpn IgG(-)	96.1	77.3	68.6	
Cpn IgG(+)	102.1	78.5	72.3	
Cpn IgG(++)	81.5	78.1	63.2	
Cpn IgA Status				<0.001
Cpn IgA(-)	100.3	78.8	71	
Cpn IgA(+)	96.3	80.6	72.6	
Cpn IgA(++)	41.1	25	25	
Pathological type				0.005
adenocarcinoma	104.6	81.6	74.1	
squamous cell carcinoma	90.7	73.6	62.6	
others	69.2	52.4	52.4	
Tumor maximum diameter				<0.001
≤ 3 cm	110.8	87.1	79.9	
3~7 cm	87.1	69.3	61	
>7 cm	50.7	43.3	36.1	
Lymph node metastasis				<0.001
Yes	64.3	55.8	40.8	
No	112.8	87.2	82.3	
Chemotherapy				0.146
Yes	95.2	76.3	66.5	
No	100.2	80.5	75.6	

expression of genes involved in inflammation, proliferation, angiogenesis, invasion, and metastasis [29–32]. *Cpn* infection can also modulate the phosphatidylinositol 3-kinase (PI3K)/Akt pathway, which is a central regulator of cell metabolism, growth, survival, and autophagy [33]. Furthermore, *Cpn* may, like *Ctr* can affect the p53 pathway, which is a tumor suppressor that controls cell cycle arrest, apoptosis, senescence, and DNA repair [34].

The molecular mechanisms by which *Cpn* infection may affect the prognosis of lung cancer patients are

complex and multifactorial. The interaction between *Cpn* infection and other risk factors, such as smoking, genetic susceptibility, and environmental exposure, may also influence the outcome of lung cancer [23, 35]. Therefore, more studies are needed to elucidate the role of *Cpn* infection in lung cancer pathogenesis and progression, and to explore the potential of *Cpn* as a biomarker or a therapeutic target for lung cancer. Detection of *Cpn* infection in lung cancer patients and treatment of those infected may help extend the survival of lung cancer patients.

We acknowledge several limitations in our study. First, the subjects were all from three hospitals in Fuzhou, Fujian Province, China, which introduces some unavoidable selection biases. Second, the information on post-treatment recurrence was insufficient, potentially affecting patient survival outcomes. Third, the number of subjects with Cpn IgA(++) was small, introducing a certain degree of chance in the results. Therefore, further prospective studies with larger cohorts are warranted to confirm these findings. In our survival analysis, we considered the impact of confounding factors on our study results and adjusted for demographic characteristics, lifestyle habits, clinical staging, and treatment methods. However, there are still confounding factors such as performance status and specific treatment backgrounds that were not accounted for. The endpoint in our analysis was limited to cancer-related death. Future analyses should aim to make more comprehensive adjustments for these confounding factors and consider additional endpoints, such as quality of life and specific causes of cancer-related deaths.

Conclusion

our study provided that *Cpn* infection was an independent risk prognostic factor in patients with lung cancer. Further prospective studies of large cohorts of patients and more research is needed to elucidate the molecular mechanism by which *Cpn* infection may affect the prognosis of lung cancer patients and to explore the potential of *Cpn* as a biomarker or a therapeutic target for lung cancer.

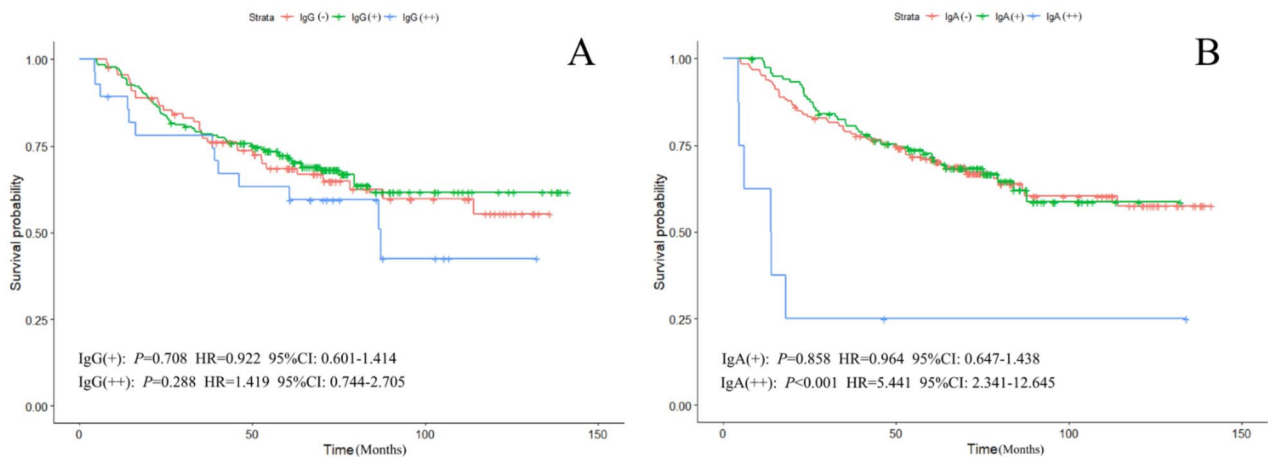


Fig. 1 Survival curves of different Chlamydia pneumoniae serum IgG and IgA infection status in lung cancer patients. **A**, Overall survival subdivided by Chlamydia pneumoniae serum IgG. **B**, Overall survival subdivided by Chlamydia pneumoniae serum IgA

Table 3 Multivariate survival analysis for OS in patients with lung cancer (Backward: LR)

Prognostic factor	HR	95% CI	P
Smoking	1.55	1.05–2.31	0.029
Lymph node metastasis	3.40	2.27–5.08	<0.001
Tumor maximum diameter			
≤ 3 cm	1.00		
3 ~ 7 cm	1.54	1.02–2.32	0.042
> 7 cm	2.36	1.11–5.01	0.026
Cpn IgA Status			
Cpn IgA(-)	1.00		
Cpn IgA(+)	0.81	0.54–1.21	0.302
Cpn IgA(++)	3.39	1.41–8.17	0.006

Variables included in the multivariate Cox regression analysis: smoking, alcohol, lymph node metastasis, Cpn IgA Status, pathological type and tumor maximum diameter

Abbreviations

Cpn	Chlamydia pneumoniae
LC	Lung cancer
LDCT	Low-dose Computed Tomography
IgG	Immunoglobulin G
IgA	Immunoglobulin A
MIF	Micro-immunofluorescence
OS	Overall survival
ROS	Reactive oxygen species
Ctr	Chlamydia trachomatis
EMT	Epithelial-mesenchymal transition
NF-Kb	Nuclear factor-kappa B
PI3K	Phosphatidylinositol 3-kinase

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12879-025-10515-3>.

Supplementary Material 1

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Not applicable.

Author contributions

FH and ML were involved in the design of the study. ZC, XX, JZ, YY, XC and CC were responsible for data collection. ML, HY and TL undertook the statistical analyses and drafted the manuscript. ML and FH contributed to revisions of the manuscript. All authors read and approved the final manuscript.

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Data availability

The data generated in this study are not publicly available due to information that could compromise patient privacy or consent but are available upon reasonable request from the corresponding author.

Declarations

Ethics approval and consent to participate

Approval was obtained from the ethics committee of Fujian Medical University (Fuzhou, China). The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. Informed consent was obtained from all individual participants included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

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